

## Chapter 21 - Ports, ACL (Access Control List), Standard ACL

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### 1 What is a Port in Networking?

A **port** is a logical communication endpoint inside a device.

Ports are entry and exit points for services 

It allows different applications and services to communicate over the network.

#### Simple Example

| Concept | Example |
|---------|---------|
|---------|---------|

|            |                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------|
| IP Address | House address  |
|------------|--------------------------------------------------------------------------------------------------|

|             |                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------|
| Port Number | Door number  |
|-------------|-------------------------------------------------------------------------------------------------|

Your house may have many rooms (applications).

The IP takes the data to the house.

The port takes it to the correct room.

Without ports 

All traffic would be confused inside the device.

#### Why Do We Use Ports?

- ✓ Allow multiple applications to run at the same time
  - ✓ Identify which service the data is for
  - ✓ Separate web, email, remote login, database traffic
  - ✓ Enable secure communication
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## Real-Life Scenario

When you open:

- Website → Uses Port 80 or 443 
- Send Email → Uses Port 25 
- Remote login → Uses Port 22 

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## 2 Port Number Ranges

| Range         | Name             | Usage                 |
|---------------|------------------|-----------------------|
| 0 – 1023      | Well-Known Ports | HTTP, DNS, FTP        |
| 1024 – 49151  | Registered Ports | Vendor apps           |
| 49152 – 65535 | Dynamic Ports    | Temporary connections |

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## 3 Common Port Numbers

| Port | Protocol | Service     |
|------|----------|-------------|
| 20   | TCP      | FTP Data    |
| 21   | TCP      | FTP Control |
| 22   | TCP      | SSH         |
| 23   | TCP      | Telnet      |
| 25   | TCP      | SMTP        |
| 53   | TCP/UDP  | DNS         |
| 67   | UDP      | DHCP Server |
| 68   | UDP      | DHCP Client |
| 69   | UDP      | TFTP        |

| Port | Protocol | Service |
|------|----------|---------|
|------|----------|---------|

|    |     |      |
|----|-----|------|
| 80 | TCP | HTTP |
|----|-----|------|

|     |     |      |
|-----|-----|------|
| 110 | TCP | POP3 |
|-----|-----|------|

|     |     |     |
|-----|-----|-----|
| 123 | UDP | NTP |
|-----|-----|-----|

|     |     |      |
|-----|-----|------|
| 143 | TCP | IMAP |
|-----|-----|------|

|     |     |      |
|-----|-----|------|
| 161 | UDP | SNMP |
|-----|-----|------|

|     |     |       |
|-----|-----|-------|
| 443 | TCP | HTTPS |
|-----|-----|-------|

|     |     |     |
|-----|-----|-----|
| 445 | TCP | SMB |
|-----|-----|-----|

|     |     |        |
|-----|-----|--------|
| 514 | UDP | Syslog |
|-----|-----|--------|

|      |     |     |
|------|-----|-----|
| 3389 | TCP | RDP |
|------|-----|-----|

|      |     |       |
|------|-----|-------|
| 3306 | TCP | MySQL |
|------|-----|-------|

|      |     |                |
|------|-----|----------------|
| 8080 | TCP | Alternate HTTP |
|------|-----|----------------|



### What is ACL (Access Control List)?

ACL is a set of rules used by routers to **permit or deny traffic**.

Think of ACL like a security guard 🚔

It checks who can enter and who must be blocked.



### Why We Use ACL?

- ✓ Improve network security
  - ✓ Restrict specific users
  - ✓ Block specific IP addresses
  - ✓ Control access to servers
  - ✓ Filter traffic based on policy
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## Where Are ACLs Applied?

### Direction Meaning

IN            Filters traffic entering interface

OUT          Filters traffic leaving interface

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## 5 Types of ACL

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### ★ 1. Standard ACL

Filters only by **Source IP**

- ✓ Simple
  - ✓ Less secure
  - ✓ Number Range: 1–99
  - ✓ Best placed near destination
- 

#### ✂ Example

```
access-list 10 deny 192.168.1.5
access-list 10 permit any
interface fa0/0
ip access-group 10 in
```

Meaning:

Block only 192.168.1.5

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### ★ 2. Extended ACL

Filters by:

- ✓ Source IP
- ✓ Destination IP
- ✓ Protocol
- ✓ Port number

Number Range: 100–199

Best placed near source

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### ✂ Example

```
access-list 101 deny tcp 192.168.1.5 0.0.0.0 10.0.0.10 0.0.0.0 eq 80
```

```
access-list 101 permit ip any any
```

Meaning:

Block HTTP from 192.168.1.5 to 10.0.0.10

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### IN vs OUT Example

If you apply:

```
ip access-group 10 out
```

It filters traffic leaving that interface.

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## STANDARD ACL FULL LAB (With OSPF)

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### Lab Topology

R1 → 2 switches → 4 PCs

R2 → 2 switches → 2 Servers

IP addressing already configured.

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### Step 1: Configure OSPF for Full Connectivity

Protocol Used: Open Shortest Path First

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#### ◆ R1 OSPF Configuration

```
router ospf 1
network 203.0.113.0 0.0.0.3 area 0
network 172.16.1.0 0.0.0.255 area 0
network 172.16.2.0 0.0.0.255 area 0
passive-interface g0/0
passive-interface g0/1
```

---

#### ◆ R2 OSPF Configuration

```
router ospf 1
network 203.0.113.0 0.0.0.3 area 0
network 192.168.1.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0
passive-interface g0/0
passive-interface g0/1
```

When adjacency forms:

%OSPF-5-ADJCHG: LOADING to FULL

That means OSPF is working correctly 

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## Step 2: Standard ACL Policies Implementation

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### Policy 1

Only PC1 (172.16.1.1) and PC3 (172.16.2.1) can access 192.168.1.0/24

#### ◆ On R2

```
access-list 10 permit 172.16.1.1
access-list 10 permit 172.16.2.1
access-list 10 deny 172.16.1.0 0.0.0.255
access-list 10 deny 172.16.2.0 0.0.0.255
access-list 10 permit any
```

```
interface g0/0
ip access-group 10 out
```

✓ Applied outbound toward 192.168.1.0 network.

---

### Policy 2

Hosts in 172.16.2.0/24 cannot access 192.168.2.0/24

#### ◆ On R2

```
access-list 10 deny 172.16.2.0 0.0.0.255
access-list 10 permit any
```

```
interface g0/1
ip access-group 10 out
```

---

### Policy 3

172.16.1.0/24 cannot access 172.16.2.0/24

#### ◆ On R1

```
access-list 10 deny 172.16.1.0 0.0.0.255
access-list 10 permit any
```

```
interface g0/1  
ip access-group 10 out
```

---

### Policy 4

172.16.2.0/24 cannot access 172.16.1.0/24

#### ◆ On R1

```
access-list 10 deny 172.16.2.0 0.0.0.255  
access-list 10 permit any
```

```
interface g0/0  
ip access-group 10 out
```

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### Verification Commands

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#### Check ACL Applied on Interface

```
show ip interface g0/0
```

Look for:

Outgoing access list is 10

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#### Check ACL Hit Counters

```
show access-lists
```

Example Output:

Standard IP access list 10

10 permit host 172.16.1.10 (5 matches)

20 permit host 172.16.2.10 (3 matches)

30 deny any (12 matches)

Matches show how many packets were filtered.



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## Important Concepts to Remember

- ✓ Standard ACL filters only source IP
  - ✓ Always add permit any at end
  - ✓ Implicit deny any exists by default
  - ✓ Apply Standard ACL near destination
  - ✓ Use correct direction (in/out)
  - ✓ Verify using show commands
- 

## Real-Life Analogy for ACL

Think of a corporate building:

Receptionist checks:

- ✓ Who are you? (Source IP)
- ✓ Which department are you going?
- ✓ Are you allowed?

If not allowed → Entry denied 

That is ACL logic.

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## Final Revision Points

- ✓ Ports are logical service doors
  - ✓ Know common port numbers
  - ✓ Understand port ranges
  - ✓ ACL improves security
  - ✓ Standard ACL → Source only
  - ✓ Extended ACL → Full filtering
  - ✓ Always verify with show commands
  - ✓ Understand IN vs OUT
-



## CHAPTER SUMMARY

**1**

### What is a Port in Networking?

A **port** is a logical communication endpoint inside a device.

- IP Address = House Address 🏠
- Port Number = Door Number 🚪

The IP brings the packet to the device.

The port sends it to the correct application (room).

Without ports ❌  
All traffic would mix together.



### Why Ports Are Important?

- ✓ Allow multiple services on one IP
- ✓ Identify application (Web, Mail, SSH, etc.)
- ✓ Enable secure communication
- ✓ Separate different types of traffic

**12****34****2**

### Port Number Ranges

| Range       | Name             | Usage              |
|-------------|------------------|--------------------|
| 0–1023      | Well-Known Ports | HTTP, FTP, DNS     |
| 1024–49151  | Registered Ports | Vendor apps        |
| 49152–65535 | Dynamic Ports    | Temporary sessions |

## 3 Important Common Port Numbers

### Port Protocol Service

20/21 TCP FTP

22 TCP SSH 

23 TCP Telnet

25 TCP SMTP 

53 TCP/UDP DNS

67/68 UDP DHCP

69 UDP TFTP

80 TCP HTTP 

110 TCP POP3

123 UDP NTP

143 TCP IMAP

161 UDP SNMP

443 TCP HTTPS 

3389 TCP RDP

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## What is ACL?

**ACL (Access Control List)** is a set of rules used to permit or deny traffic.

Think of ACL as a **security guard** 

It checks:

- Who is sending traffic (Source)
- Where it is going (Destination – extended ACL)
- What type of traffic (Protocol/Port – extended ACL)

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## Why Use ACL?

- ✓ Improve security
  - ✓ Block unwanted users
  - ✓ Restrict server access
  - ✓ Control traffic flow
  - ✓ Implement company policy
- 

## 5 Types of ACL

### ★ Standard ACL

- ✓ Filters by Source IP only
- ✓ Number range: 1–99
- ✓ Less secure
- ✓ Best placed near destination

Example:

```
access-list 10 deny 192.168.1.5
access-list 10 permit any
interface fa0/0
ip access-group 10 in
```

Blocks only that source IP.

---

### ★ Extended ACL

- ✓ Filters by:
  - Source IP
  - Destination IP
  - Protocol
  - Port number

✓ Number range: 100–199

✓ Best placed near source

Example:

```
access-list 101 deny tcp 192.168.1.5 0.0.0.0 10.0.0.10 0.0.0.0 eq 80
```

```
access-list 101 permit ip any any
```

Blocks HTTP traffic only.

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### IN vs OUT

- IN → Filters traffic entering interface
  - OUT → Filters traffic leaving interface
- 

### Important ACL Concepts

- ✓ ACL works top-to-bottom
  - ✓ First match wins
  - ✓ Implicit deny any at end
  - ✓ Always add permit any (if required)
  - ✓ Verify using show commands
- 

### Standard ACL + OSPF Lab Summary

- 1 Configure OSPF for full connectivity
- 2 Apply Standard ACL policies
- 3 Apply near destination
- 4 Verify using:

show access-lists

show ip interface

Hit counter shows how many packets matched rule.

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## FINAL CONCLUSION

This chapter teaches:

- ✓ How applications communicate (Ports)
- ✓ How services are identified
- ✓ How to secure networks using ACL
- ✓ Difference between Standard & Extended ACL
- ✓ How to apply ACL in real scenarios

👉 Ports = Service identification

👉 ACL = Traffic control & security

Without ACL → Network is open 🚨

With ACL → Controlled & Secure 🔒



## Q & A

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### 1 What is a Port?

A port is a logical endpoint that identifies a specific service or application on a device.

It allows multiple services (web, email, SSH) to run on one IP.

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### 2 What is the difference between IP address and Port?

IP = Identifies device

Port = Identifies application

Together they form socket:

IP + Port = Communication session

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### 3 What are well-known ports?

Ports 0–1023 reserved for common services like HTTP (80), HTTPS (443), SSH (22).

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### 4 What is ACL?

ACL is a set of filtering rules applied on routers to permit or deny traffic.

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### 5 Why do we use ACL?

To control traffic, increase security, and block unauthorized access.

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### 6 Difference between Standard and Extended ACL?

| Standard         | Extended                    |
|------------------|-----------------------------|
| Source only      | Source + Destination + Port |
| 1–99             | 100–199                     |
| Near destination | Near source                 |
| Basic filtering  | Advanced filtering          |

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### 7 What is implicit deny?

Every ACL ends with:

deny any

Even if not written.

If no match → packet dropped ❌

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### 8 Why Standard ACL placed near destination?

Because it filters only source.

If placed near source → may block valid traffic unnecessarily.

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### 9 How ACL processes rules?

Top to bottom.

First match stops processing.

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### 10 What is wildcard mask?

Wildcard mask is inverse of subnet mask.

Example:

Subnet mask: 255.255.255.0

Wildcard: 0.0.0.255

Used in ACL to match IP ranges.

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### 1 1 What happens if permit any not added?

Implicit deny blocks all remaining traffic.

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### 1 2 Can we apply multiple ACLs on same interface?

Only:

- One inbound per protocol
  - One outbound per protocol
- 

### 1 3 What happens if ACL applied wrong direction?

Traffic may not be filtered correctly.

Example:

If filtering outbound but traffic coming inbound → rule ineffective.

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### 1 4 How to verify ACL?

show access-lists

show ip interface

Hit count shows packet matches.

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### 1 5 What is difference between eq, gt, lt in extended ACL?

eq → equal to port

gt → greater than

lt → less than

Used for port filtering.

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### 1 6 Scenario: Block HTTP but allow HTTPS

access-list 101 deny tcp any any eq 80

access-list 101 permit tcp any any eq 443

access-list 101 permit ip any any

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### 1 7 Why ACL important in enterprise?

Prevents unauthorized access to servers.

Improves security compliance.

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### 1 8 How to remove ACL?

no access-list 10

Remove from interface:

no ip access-group 10 in

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### 1 What is named ACL?

Instead of numbers:

```
ip access-list standard BLOCK_USERS
```

More readable & flexible.

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### 2 Difference between Standard ACL & Firewall?

ACL:

- Basic filtering
- Stateless

Firewall:

- Stateful inspection
  - Application awareness
  - Deep packet inspection
- 



### PRACTICAL SCENARIO QUESTIONS

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#### 2 Block one PC from server network

Use Standard ACL near server network.

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#### 2 Block only FTP traffic

Use Extended ACL with port 21.

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#### 2 How to check which ACL applied?

```
show ip interface g0/0
```

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**2 4 What happens if ACL not applied to interface?**

It exists in config but does nothing.

ACL must be applied to interface to work.

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**2 5 Why ACL hit count important?**

Shows how many packets matched rule.

Useful for troubleshooting.

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KNM